

DATA QUALITY REPORT

NexaTel Customer Churn Analytics

The purpose of this report is to assess the quality, consistency, validity, and analytical readiness of the NexaTel data before EDA, churn KPI calculations, SQL analysis, and Power BI reporting.

The project contains approximately 757,000 records across 24 interconnected tables, covering customers, subscriptions, billing, payments, usage, network quality, complaints, support, and campaigns.

Executive Summary

Overall, the NexaTel data contains several identifiable data-quality issues that require cleaning before churn analysis. The major issues include duplicate records, missing customer information, invalid customer attributes, billing inconsistencies, broken foreign-key relationships, and usage outliers.

The Customers data contains 76 duplicate records, along with invalid phone numbers, impossible ages, missing annual income, and inconsistent customer information. The Billing data contains 344 duplicate invoices, 229 negative invoice amounts, 229 future billing dates, and 1,149 incorrect GST records.

The assessment also identified 184 broken Payment-to-Billing relationships, 276 duplicate payments, 41 null Subscription foreign keys, 102 invalid Subscription categories, and 171 usage outliers.

These issues should be resolved or appropriately documented before calculating churn, ARPU, recurring revenue, customer lifetime value, and other business KPIs.

Data Quality Summary

Area	Key Finding	Assessment
Customers	76 duplicate records	Requires Cleaning
Customers	228 invalid phone numbers	Requires Correction
Customers	38 impossible ages	Requires Correction
Customers	950 missing annual income values	Requires Treatment
Billing	344 duplicate invoices	Requires Cleaning
Billing	229 negative invoice amounts	Requires Investigation
Billing	229 future billing dates	Requires Investigation
Billing	1,149 incorrect GST records	Requires Correction
Payments	184 broken invoice references	Integrity Issue
Payments	276 duplicate records	Requires Cleaning
Subscriptions	41 null foreign keys	Integrity Issue
Subscriptions	102 invalid categories	Requires Correction
Usage Data	171 outlier values	Requires Review

The project specifically requires checks for nulls, duplicates, date and numeric types, usage outliers, negative billing amounts, and customer-level referential integrity.

Key Findings

Customer Data: Duplicate customers, invalid phone numbers, impossible ages, missing values, and inconsistent formats were identified. These issues can affect customer counts, segmentation, and churn-rate calculations.

Billing & Payments: Duplicate invoices, negative amounts, future billing dates, incorrect GST values, duplicate payments, and broken invoice references were identified. These issues can affect revenue-related KPIs.

Subscriptions & Usage: Null foreign keys and invalid subscription categories were detected, along with 171 data-usage outliers requiring further validation.

Complaints: The notebook identified 9,236 Resolved and 7,348 Unresolved complaints, meaning a substantial share of complaints remain unresolved. This is important because the project specifically investigates the relationship between unresolved complaints and churn

Issues Requiring Attention

Duplicate Records: Remove confirmed duplicate customers, invoices, and payments to prevent inflated counts and incorrect KPI calculations.

Invalid & Missing Customer Data: Validate phone numbers, ages, PIN codes, emails, and other customer attributes. Missing values should be handled according to their business meaning.

Billing Issues: Investigate negative invoice amounts, future billing dates, and incorrect GST values before performing revenue analysis.

Referential Integrity: Resolve broken Payment-to-Billing references and missing Subscription foreign keys before joining related tables.

Data Types: Standardize date fields using the expected DD-MM-YYYY format and ensure financial and usage measures are numeric, as required by the project specification.

Final Assessment

The NexaTel datasets contain multiple quality issues, but they can be addressed through systematic cleaning, validation, and standardization.

The main priorities are removing duplicates, correcting invalid customer information, resolving billing inconsistencies, validating foreign-key relationships, and reviewing outliers.

Overall Data Readiness Status: Suitable for Analysis After Identified Data-Quality Issues Are Addressed